

HIMANSHU KUMAR PATHAK

AI/ML Undergraduate

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EDUCATION

• Vellore Institute of Technology, Bhopal

BTech, Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Sept 2023 - Present

CGPA: 8.58

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, SQL
- **Machine Learning & Data Science:** Scikit-learn, SHAP, Random Forest, XGBoost, LightGBM, CatBoost
- **Generative AI & RAG:** LangChain, ChromaDB, RAG, Embeddings
- **Tools & Technologies:** Git, GitHub
- **Core Subjects:** Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming

PROJECTS

• Proposia – AI Sales Proposal Copilot

[GitHub](#)

Tools: Python, Streamlit, LangChain, ChromaDB, Sentence Transformers, Groq

- Built a **RAG-based** sales proposal copilot that retrieves relevant company capabilities, case studies, technology stack, and pricing to generate client-specific proposals.
- Developed an end-to-end knowledge ingestion pipeline for **PDF, Markdown, CSV, and JSON** documents with processing, chunking, metadata management, and **384-dimensional embeddings**.
- Implemented semantic retrieval using **ChromaDB** and **cosine similarity**, processing both text and PDF-based RFPs to retrieve context relevant to client requirements.
- Integrated **Groq LLM** inference and a **Streamlit** interface for grounded proposal generation and conversational refinement of generated proposals.

• CareerPilot AI – Explainable AI Career Path Intelligence System

[GitHub](#)

Tools: Python, Scikit-learn, Random Forest, SHAP, Streamlit

- Built an end-to-end explainable AI **career path intelligence platform** that predicts suitable career paths from 37 user attributes across **20 career categories**.
- Developed a **synthetic-data generation** and validation pipeline with 20,000 balanced candidate profiles.
- Trained and compared **Random Forest, XGBoost, LightGBM, and CatBoost** models, achieving **99.78%** test accuracy and **99.59%** cross-validation accuracy after hyperparameter tuning.
- Implemented **SHAP-based explainability**, confidence scoring, recommendation orchestration, skill-gap analysis, and personalized learning roadmap generation using a structured career knowledge base.

EXTRACURRICULAR ACTIVITIES

• Lead, Technical Team – AdVITya'26 Annual Sports Fest

Vellore Institute of Technology, Bhopal

- Coordinated **300+ student volunteers** and managed technical operations across multiple committees during the annual sports fest.
- Oversaw participant registration and operational logistics for **1,500+ participants**.

ADDITIONAL INFORMATION

- Python for Data Science, AI Development – IBM (Coursera)
- Applied Machine Learning in Python — University of Michigan (Coursera)
- International Mathematics Olympiad: School Rank 2
- Community Engagement: Participated in an **NSS Plantation Drive** promoting environmental awareness.
- Language: English & Hindi